5. Local Exhaust Ventilation

The objective of a local exhaust ventilation system is to remove the contaminant as it is generated at the source. Gases and vapors are controlled by controlling the air in which they are contained. Particulates of industrial hygiene significance (i.e., d < 10μm) are controlled by controlling the air in which they exist. These factors were illustrated in some detail in Chapter 2 of this section. Special procedures are required to control large particles that are generated at the source. These particles are controlled for other than health purposes.

Components of a Local Exhaust System

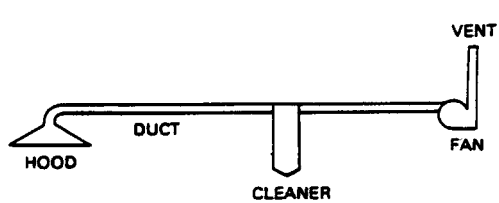
The initial opening through which contaminated air enters a local exhaust system is called the hood. The term hood is used generically for any opening whether it is specifically designed or consists of simply the open end of a round or rectangular duct section. Many hoods are specifically designed and located to meet the requirements of the operation and the contaminant being generated. Later in this chapter, the general design characteristics that describe different types of specially constructed hoods will be discussed.

After the contaminated air has entered the hood, it flows through a duct system which directs the flow of contaminated air and prevents mixing of this air with the workroom atmosphere. Branches may exist within the duct to join separate local systems into one single exhaust system.

The third component of a local exhaust system is a method for cleaning the air. It is often necessary to remove the contaminant from the air before exhausting the air into the atmosphere to prevent hazardous materials from entering the breathing zone of individuals in the community surrounding the

Figure 2.5.1

Local exhaust system components.



plant. Various types of air cleaners are used: filters, precipitators, and cyclones, among others. The type of cleaner used is selected for the particular characteristics of the contaminant that is being controlled. The design and operation of air cleaners will be covered in a later chapter.

The next component of a local exhaust system is the air mover or fan. The fan provides a source of air movement in a local exhaust system. Infrequently, natural draft is used as a source of air movement. This is particularly the case in hot processes that exhaust upward through a hood and outside the work area through vents on the roof. The fan to be used in a local exhaust system is sized to meet the specific air movement characteristics of the exhaust system. The desired rate of flow in cubic feet per minute and the system's static pressure are the major considerations when determining the size of fan to be used. A more complete discussion of the various types of air movers that are available and the appropriate methods for sizing these air movers will be discussed in a later chapter.

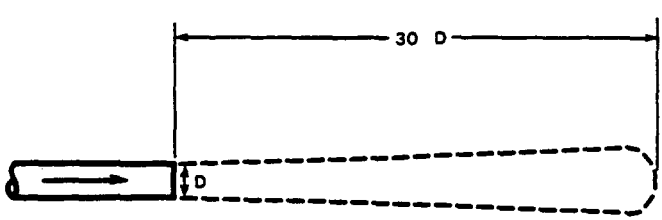
The final component of a local exhaust system is the outlet or vent. Often the outlet or vent for contaminated air is on the roof or through exterior stacks. In either case, care must be taken to avoid locating the outlet near any potential re-entry point to the plant to avoid recontamination of the workplace air. If an air cleaner is present, it is desirable to attach monitoring devices to assure operation of the air cleaner so that contaminated air is not exhausted in the case of equipment failure.

Blow Versus Exhaust

It is important that the reader understand the difference between the effects of blowing air from a supply system and removing air through an exhaust system. When air is supplied through a duct, it retains its directional effect for some distance beyond the end of the duct itself. At a distance of 30 diameters from the outlet, the velocity of the air is 10% of

Figure 2.5.2

Blowing air.



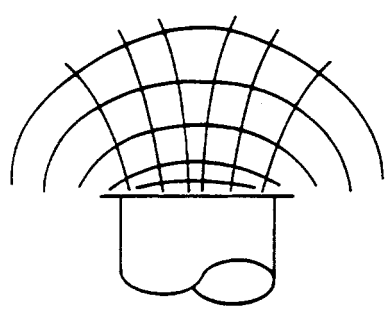
the velocity at the outlet. The diameter of the air jet increases with increasing distance from the nozzle but still retains essentially the same form for a significant distance.

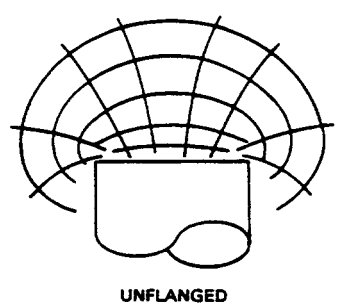
As an illustration of this effect, consider the use of a common garden hose squirting water. Depending on the force of water pressure at the hose and the orifice for which the hose nozzle is set, water can be caused to travel for a significant distance from the nozzle. The water spray at a distance away from the nozzle is obviously wider than it is immediately after leaving the nozzle. However, the directional effect and the cross-sectional shape of the spray remain essentially the same.

On the other hand, consider what happens when water is collected using a wet vacuum cleaner. If a small piece of paper is placed a short distance away from the suction end of the wet vacuum, it will not be very easily gathered with the water. The same thing happens when air is exhausted through a local exhaust system. By changing the direction of the air flow through the system, we do not obtain similar velocity patterns. The velocity of the air through a local exhaust system is 10% of the inlet velocity at a distance of only one diameter from the inlet. This is illustrated by looking at the air contours that are involved in exhaust through an opening.

Figure 2.5.3

Exhausting air.





When air is exhausted through an opening, it is gathered equally from all directions around the opening. This includes the area behind the opening itself. Thus, the cross-sectional area of air flow approximates a spherical form rather than the conical form that is obtained when air is blown out of a supply system. By placing a flange around the exhaust opening, it is possible to reduce the air contour from the large spherical contour to that of a hemisphere and, as a result, increase the velocity of air at a given distance from the opening. This basic principle is used in designing exhaust hoods. The closer the exhaust hood is to the source and the less uncontaminated air it gathers, the more efficient the capture of the hood will be.

Rate of Flow in a Hood

Air that enters a local exhaust system enters at the hood face (the cross-sectional area of the hood at its opening). At this point, air has a given velocity that is called the face velocity. The face velocity is approximately equal over all points on the face of the hood if the hood has been properly designed. The rate of flow for a hood is given as the face velocity times the cross-sectional area of the hood opening.

$$Q = vA \quad (2.3.4)$$

where

Q = the rate of flow in cfm
v = the face velocity of the hood in fpm
A = the area of the hood face in ft^2

Example. To illustrate the use of this formula, assume a face velocity is measured at 300 fpm on a circular flanged hood of diameter 10 inches. The rate of air flow in cubic feet per minute that is present in this system, assuming standard conditions, can be calculated as follows:

$$Q = vA \quad (2.3.4)$$
$$Q = 300 \text{ fpm} \times 0.5454 \text{ ft}^2$$
$$Q = 164 \text{ cfm}$$

The Application and Advantage of Local Exhaust Ventilation

One of the major advantages of a local exhaust system is the fact that the contaminant is removed at the source. This removal is necessary when working in areas where highly toxic contaminants are present. Though dilution ventilation can be used to control such contaminants, quite often the rate of flow that is required is too high to be practical as a method of control. In these cases, local exhaust ventilation systems are most appropriate.

Local exhaust ventilation systems require lower air volumes than are required for dilution ventilation. As a result, a lower cost of operation in terms of tempering the workplace air is experienced. Lower air cleaning costs are also encountered. The initial expense in installing the necessary duct can be partially offset by the fact that the air moving devices can be sized smaller to move the lower air volume required.

The air volume that is moved through the local exhaust ventilation system is dependent upon the required capture velocity for the contaminant. Capture velocity is the velocity that is necessary to capture the contaminant at its farthest possible distance from the hood. From the previous discussions concerning the characteristics of exhaust air contours, it is obvious that this distance cannot be very great. As the distance from the source to the hood is increased, the rate of flow of the system must also be increased in order to obtain the desired capture velocity at the source.

When designing and installing a local exhaust ventilation system, it is important to consider the presence of any air currents that might cause the contaminant to be moved away from the hood. Potential sources of external air motion that can result in inadequate contaminant capture in a local exhaust system include:

1. Currents caused by motion in the plant.
2. Currents caused by general ventilation systems.
3. Thermal draft.
4. Motion imparted to the contaminant air by the process.
5. Entrainment of contaminated air by motion of large particles that are also generated by the process.
6. Motion of the air caused by indirect mechanical action of the processing equipment.

The reader should remember that gases, vapors, and particulates that are of industrial hygiene significance follow the air currents and tend not to settle out or have projected motion. Thus, to remove these contaminants, control of the air flow is necessary.

General Categories of Local Exhaust Ventilation Hoods

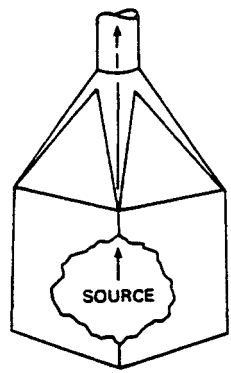
Enclosing Hoods--Total Enclosures. The most efficient type of exhaust ventilation hood that can be constructed is a hood that encloses the entire operation. Openings in this enclosure are minimized with doors being used for access to the contaminant generation zone. Obviously, if no openings are required, and the enclosure can be made air-tight, then exhaust of this area is not necessary since there is no chance for the contaminant to escape into the workroom atmosphere.

The enclosing hood should be kept under negative pressure to create an inflow of air from the workroom atmosphere into the contamination zone through any openings that do exist. The capture velocity at the source of contaminant generation must be sufficient to overcome any air movement inside the enclosure itself. The effect of outside air movement is minimized in a total enclosure.

Generally, an enclosing hood is used where the process does not require that workers be present within the contaminant generation zone. Process and equipment limitations do not always allow for use of an enclosing hood. Openings may be necessary for material to enter and exit the process or

Figure 2.5.4

Total enclosure.



machinery. In some cases, it may be necessary for the worker to interact continuously or at intervals with the process. In these cases, openings are required for the worker's access, and a total enclosing hood cannot be applied.

Enclosing hoods have been applied in many situations. Some examples of enclosing hoods are presented in the Industrial Ventilation Manual, Section 5. If it is necessary to design a total enclosure for an exhaust ventilation system, these examples should be studied. Empirical data for the designs presented have been gathered for such things as the location of supply and exhaust vents, the velocity of the air, and the rate of flow that is necessary to control a given contaminant. Enclosing hoods have been used in such areas as abrasive blasting, crucible melting furnaces, foundry shakeouts, mixing and mulling machines, processes generating highly toxic contaminants, and screening applications. Wherever possible, the enclosing hood is the most appropriate type of exhaust hood to apply because of its inherent efficiency in capturing generated contaminants.

Enclosing Hoods--Booths and Tunnels. Where there is a necessity for access either on one side or two sides of a process, the most desirable type of enclosing hood to use is the booth or tunnel.

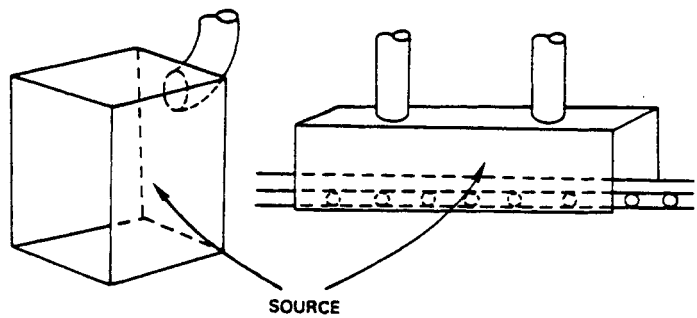
The face velocity of such an enclosure is determined at the opening of the tunnel or booth. The face velocity must be sufficient to eliminate the escape of air from the enclosure and to eliminate effects of external movement on the process within the enclosure.

The worker may work either inside or outside the booth. If the worker is internal, the work position should be located between the face of the booth and the exhaust inlet.

It is obvious that a booth or tunnel will require higher exhaust rates than will a total enclosure. Thus, the booth or tunnel is less efficient.

Figure 2.5.5

Booth and tunnel hoods.



However, certain applications require the use of such an exhaust hood. Examples of the applications of the booth or tunnel include laboratory hoods, abrasive cut-off saws, continuous buffing where a tunnel is used, material movement where entry and exit points are necessary, grinding, paint spraying, and welding.

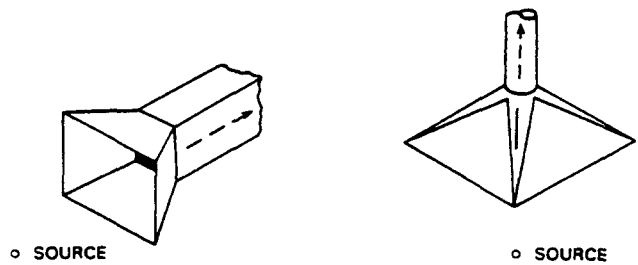
Rate of Flow for Enclosing Hoods. The air flow rate for an enclosing hood is determined by the required face velocity at any of the openings that are necessary to control for such things as internally induced air movement--including thermal effects, mechanical movements, process material movement, and worker movement--and externally induced air movement that may be caused by general ventilation systems, natural air currents, and movement of equipment and workers external to the process. The face velocity at any opening must be sufficient to overcome both the internal and external air movement. Thus, to determine the air flow rate for an enclosing hood, one should determine the velocity that is necessary at the face of the hood. The area of opening in square feet comprises the face area of the hood. Then, using $Q = vA$, the required rate of flow in cfm can be calculated.

Exterior Hoods. Where it is not possible to use a total enclosure or a partial enclosure such as a booth or tunnel, the exterior hood can be applied. Exterior hoods are hoods that are adjacent to the sources of contamination. The operation itself is exterior to the boundary of the hood face. For such a hood, the capture velocity is created at the source to overcome any movement and capture the contaminant. Capture velocity must be such as to overcome any exterior air currents, overcome any process-induced motion such as projected motion, and move the contaminant from its most distant location to the hood face.

Exterior hoods are inefficient when compared to enclosure-type hoods. Air enters the hood from all directions, and the area between the source and hood is subject to the effects of external air movement. External hoods are applied where it is not possible to use the more efficient type of hoods.

Figure 2.5.6

Exterior hood.



The hood itself may be located in the path of contaminant generation in order to improve the capture of contaminants.

There are four general categories of exterior hoods. Two categories are by type of face opening, and two are by location of the hood relative to the source.

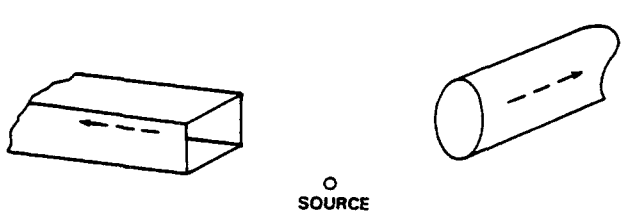
1. Type of Opening
   A. Round or rectangular duct openings
   B. Slot openings

2. Location Relative to Source
   A. Canopy hoods
   B. Receiving hoods

The round or rectangular duct opening is the simplest type of hood. It is also one of the most inefficient. The operation generating the contaminant must be close to the hood face in order to provide adequate capture velocity. The hood itself can be attached to a flexible duct so that it can be moved to capture sources of contamination that may be generated at different locations in the work area. As has been previously discussed, a flange or taper can be added to the round or rectangular opening hood to increase its efficiency.

Figure 2.5.7

Round or rectangular hoods.



The round or rectangular duct opening hood can be used in many applications. Examples of its use include welding benches, surface grinding, and hand saws. In addition, it can be applied to various portable and hand tools when flexible duct or rubber hose is used.

The second category of exterior hood is the slot hood (Figure 2.5.8). Slot hoods are defined as hoods with openings whose width-to-length ratio is less than 0.2. The advantage of the slot hood is that it maintains an air flow over a large surface plane. Quite often the slot hood is used along the sides of large open tanks. If the distance for capture is great (the width of the tank is too large to apply adequate velocity at the midpoint of the tank), the slot opening is used in combination with a blowing supply slot. This type of ventilation system is called a "push-pull" ventilation system and creates a plane of air that acts as a barrier to any contaminant generated in the tank.

The slot hood is less efficient than a round or rectangular opening when it is necessary to remove air from a given point. Obviously, a series of discrete points of contamination would best be exhausted (not considering economic circumstances) by applying a series of round or rectangular hoods at each point source. Where the source of contamination is continuous over the entire length of the slot, this method is best.

The slot hood has been applied in a number of areas. There are slot hoods for belt wiping, slot hoods on welding benches where any point may be used for the actual weld, and slot hoods on degreasing tanks, other open surface tanks, and process tables where any point may be used for the process.

The third type of exterior hood is the canopy hood. Canopy hoods are overhead openings that are generally used for hot processes which cause thermal draft. Canopy hoods can also be used for cold processes where, for example, it is necessary to control evaporation from an open tank. Sides or baffles can improve the operation of a canopy hood by approximating the design of an enclosed hood or booth.

Figure 2.5.8

Slot hood.

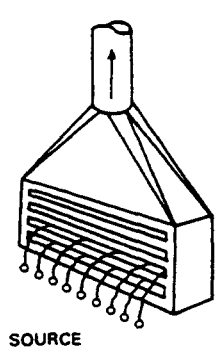
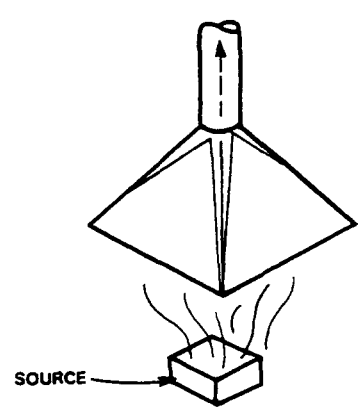


Figure 2.5.9

Canopy hood.

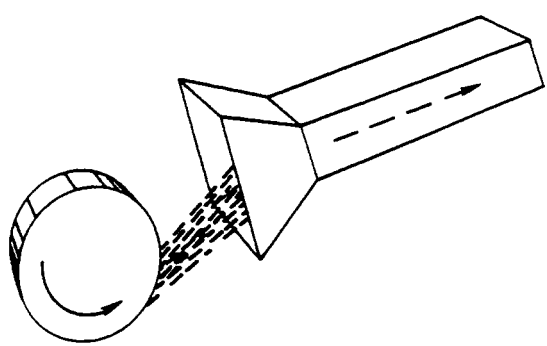


Applications of the canopy hood include die casting melting furnaces and the open surface tank.

The fourth type of external exhaust hood is the receiving hood. The receiving hood is actually one of the above types of hoods where the exhaust port itself is located in the path of contaminant generation. Thus, it is located at a point that receives any process-induced contamination. The canopy hood is a type of receiving hood where the process-induced motion caused by thermal effects is upward. The receiving hood takes advantage of any directional inertia of the contaminant, thus lowering the necessary capture velocity that must be applied. Examples of receiving hoods include hoods used for grinding wheels, polishing wheels, and hoods used for various types of table saws or band saws.

Figure 2.5.10

Receiving hood.



Summary

Local exhaust systems are used to capture the contaminant at the source before entry into the workroom air. Local exhaust systems generally include as components the hood, duct, an air cleaner, an air mover, and a vent or outlet. Local exhaust systems must provide the necessary control of the contaminant-laden air at the source. Because air is drawn into the hood from all points around the hood face, the hood must be relatively close to the source. In addition, the hood must be designed to be as efficient as possible. This involves enclosing as much of the source of the contamination as is possible without affecting the process operation. The major categories of hoods are the enclosing hood, partial enclosing hoods (booth or tunnel), and the exterior hood.